Experiment No 9

Aim: **Graphical Simulation of Network with routing protocol and traffic**

Lab Outcomes:

After completing this experiment, students will be able to:

1. Understand how to perform graphical network simulations using **NS-2**.
2. Learn to configure nodes, links, and network topology in a simulation script.
3. Implement and analyze routing protocols such as **AODV** in the simulated network.
4. Generate and configure traffic sources like **UDP** and **TCP**.
5. Visualize packet transmission and routing behavior using the **NAM** tool.
6. Analyze the flow of data packets and observe network performance in a graphical environment

Theory:

1. Network Simulation in NS-2

NS-2 is an open-source discrete event simulator widely used for research and educational purposes in computer networking. It allows users to simulate different network protocols, traffic models, and topologies. In NS-2, the main simulation logic is written using **Tcl**, while the core components of the simulator are implemented in **C++**. This dual-language architecture enables flexible simulation configuration along with efficient execution.

2. Routing Protocols in Network Simulation

Routing protocols determine the path that data packets take from the source to the destination node in a network. In NS-2 simulations, routing protocols such as **AODV** are commonly used for wireless and mobile ad-hoc networks. AODV is a reactive routing protocol that establishes routes only when required, which helps reduce overhead in dynamic networks. By simulating routing protocols, researchers and students can analyze route discovery, packet forwarding, and network efficiency.

3. Traffic Generation

Traffic generation is an important part of network simulation because it represents real data communication between nodes. NS-2 supports several traffic types, including **TCP** and **UDP** traffic. TCP provides reliable data transmission with congestion control, while UDP is connectionless and faster but does not guarantee delivery. Traffic generators such as Constant Bit Rate (CBR) can be attached to nodes to simulate continuous data transmission in the network.

4. Graphical Visualization using NAM

The simulation results in NS-2 can be visualized using **NAM**. NAM provides a graphical interface that displays nodes, links, packet movement, and routing paths during the simulation. This visualization helps users easily understand how packets travel through the network, how routing decisions are made, and how traffic flows between nodes.

Code:

```
# Create a simulator object

set ns [new Simulator]


# Define different colors

# for data flows (for NAM)

$ns color 1 Blue

$ns color 2 Red


# Open the NAM trace file

set nf [open out.nam w]

$ns namtrace-all $nf


# Define a 'finish' procedure

proc finish {} {

    global ns nf

    $ns flush-trace
```

```
    # Close the NAM trace file

    close $nf



    # Execute NAM on the trace file

    exec nam out.nam &

    exit 0

}



# Create five nodes

set n0 [$ns node]

set n1 [$ns node]

set n2 [$ns node]   ;# central node

set n3 [$ns node]

set n4 [$ns node]



# Create links (Star topology with n2 as center)

$ns duplex-link $n0 $n2 2Mb 10ms DropTail

$ns duplex-link $n1 $n2 2Mb 10ms DropTail
```

```
$ns duplex-link $n3 $n2 1.7Mb 20ms DropTail

$ns duplex-link $n4 $n2 1.5Mb 15ms DropTail


# Set Queue Size

$ns queue-limit $n2 $n3 10


# Give node position (for NAM)

$ns duplex-link-op $n0 $n2 orient left-up

$ns duplex-link-op $n1 $n2 orient left-down

$ns duplex-link-op $n3 $n2 orient right-up

$ns duplex-link-op $n4 $n2 orient right-down


# Monitor the queue

$ns duplex-link-op $n2 $n3 queuePos 0.5


# Setup a TCP connection (n0 → n3)

set tcp [new Agent/TCP]
```

```
$tcp set class_ 2

$ns attach-agent $n0 $tcp


set sink [new Agent/TCPSink]

$ns attach-agent $n3 $sink


$ns connect $tcp $sink

$tcp set fid_ 1



# Setup a FTP over TCP connection

set ftp [new Application/FTP]

$ftp attach-agent $tcp

$ftp set type_ FTP



# Setup a UDP connection (n1 → n4)

set udp [new Agent/UDP]
```

```
$ns attach-agent $n1 $udp

set null [new Agent/Null]

$ns attach-agent $n4 $null

$ns connect $udp $null

$udp set fid_ 2

# Setup a CBR over UDP connection

set cbr [new Application/Traffic/CBR]

$cbr attach-agent $udp

$cbr set type_ CBR

$cbr set packet_size_ 1000

$cbr set rate_ 1mb

$cbr set random_ false
```

```
# Schedule events

$ns at 0.1 "$cbr start"

$ns at 1.0 "$ftp start"

$ns at 4.0 "$ftp stop"

$ns at 4.5 "$cbr stop"


# Detach agents

$ns at 4.5 "$ns detach-agent $n0 $tcp ; $ns detach-agent $n3 $sink"


# Call finish procedure

$ns at 5.0 "finish"


# Print CBR packet size and interval

puts "CBR packet size = [$cbr set packet_size_]"

puts "CBR interval = [$cbr set interval_]"


# Run the simulation

$ns run
```

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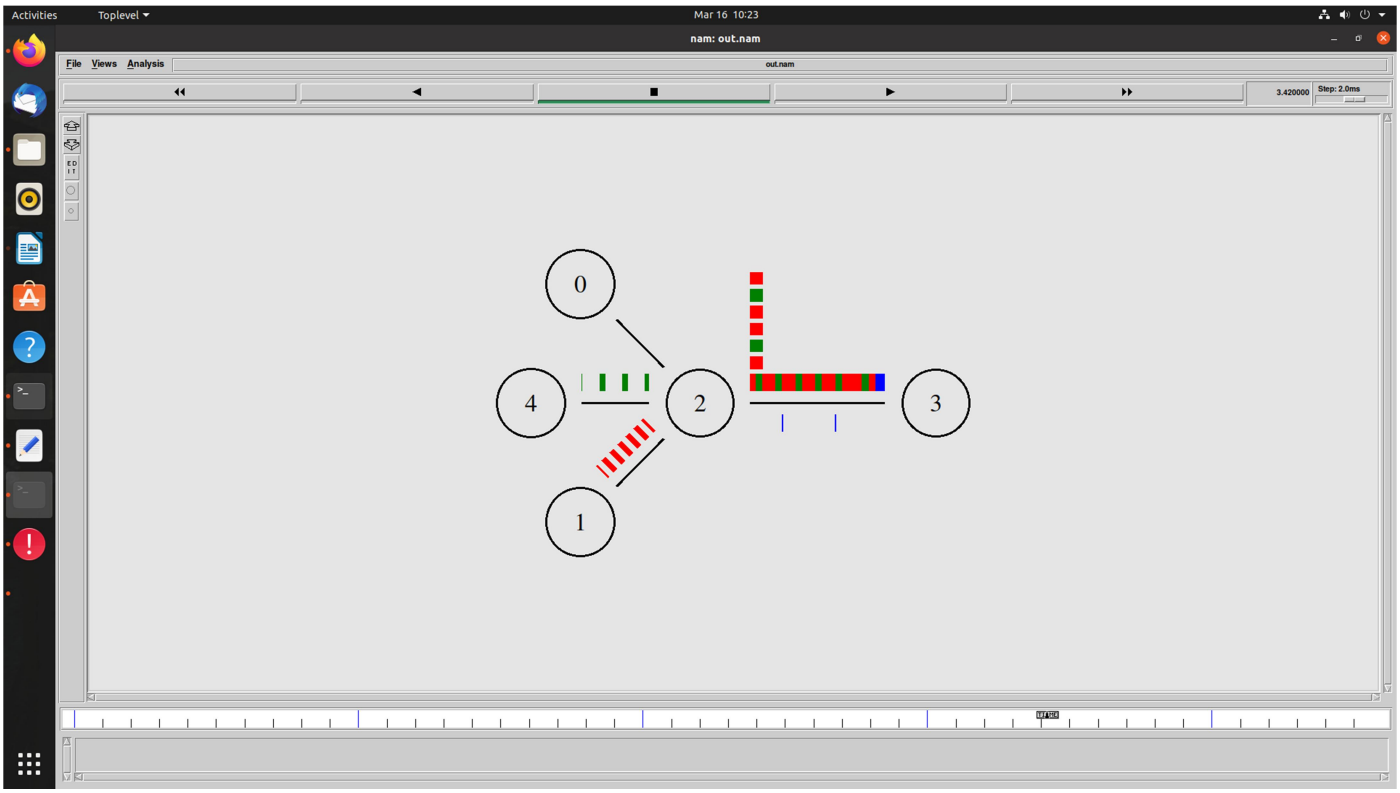
Output:



Conclusion:

The graphical simulation of a network using **NS-2** helps in understanding the behavior of routing protocols and traffic flow in a network environment. By implementing routing protocols like **AODV** and generating traffic such as **TCP** or **UDP**, students can observe packet transmission and routing decisions visually through **NAM**. This experiment provides practical insight into network behavior and performance analysis.